

EPE Progress Demo Status - Backend Section

Date: 22 September 2025 Team Member: Byron Norval - 21444758 Role: Backend Game Logic Developer

Completed Tasks (Done Status)

Core Rules Engine Implementation				
ID	Task	Completed?	Tested?	Commented?
ENG-01	6×6 Los Alamos board representation and piece placement	Yes	Yes	Yes
ENG-02	FEN string parsing and generation (authoritative)	Yes	Yes	Yes
ENG-03	Legal move generation per piece type	Yes	Yes	Yes
ENG-04	Apply moves and update game state (toMove, ply, clocks)	Yes	Yes	Yes
ENG-05	Variant-specific rules enforcement (no castling, modified promotion)	Yes	Yes	Yes
Advanced Rules Engine Features				
ENG-06	Check detection implementation	Yes	Yes	Yes
ENG-07	Checkmate & stalemate conditions	Yes	Yes	Yes
ENG-08	Threefold repetition & 50-move rule draws	Yes	Yes	Yes
ENG-09	SAN generation for all legal moves with flags	Yes	Yes	Yes
ENG-10	Illegal move rejection and validation	Yes	Yes	Yes
Complete AI Bot Implementation				
AI-01	Stub Bot API & integration with engine	Yes	Yes	Yes
AI-02	Bot Level 0 (pure random move selection)	Yes	Yes	Yes
AI-03	Bot Level 1 (1-ply evaluation with material balance)	Yes	Yes	Yes
AI-04	Bot Level 2 (2-ply minimax with basic evaluation)	Yes	Yes	Yes
AI-05	Bot Level 3 (deeper minimax/alpha-beta with heuristics)	Yes	Yes	Yes
AI-06	Time-capped search respecting msCap limits	Yes	Yes	Yes
AI-07	Bot audit logging for reproducibility	Yes	Yes	Yes

Code Quality & Integration

Unit Tests: test-rules-engine.js (move validation, checkmate/stalemate), test-ai-bot.js (L0-L3 timing), test-ai-bot-stress.js (performance), test-board-validator.js, test-piece-movement.js, test-file-loading.js

Code Structure: backend/src/engine/ (index.js, board-validator.js, constants.js, game-state-checker.js, piece-movement.js), backend/src/ai-bot/ (index.js, evaluator.js, strategies/)

Performance: Move Validation < 5ms, Legal Move Generation < 10ms, Bot L0 < 50ms, Test Coverage 85%

Demo Readiness & Metrics

Functional: Human vs Human with complete validation, Check/checkmate/stalemate detection, FEN round-trip, All Bot Levels (L0 Random, L1 Greedy 1-ply, L2 2-ply minimax, L3 Enhanced), Time-capped responses, Threefold repetition & 50-move rule, Deterministic behavior with seedable RNG

Code Metrics: Rules Engine 650 lines, AI Bot 420 lines, Unit Tests 580 lines, **Total 1,650 lines**

Status: 100% of assigned tasks completed , now needing to ensure smooth intergration.

EPE Progress Demo Status - Database Section

Date: 22 September 2025 **Team Member:** Arno Meyer - 23530848 **Role:** Database & Persistence Developer

Completed Tasks (Done Status)

Core Database Schema Implementation				
ID	Task	Completed?	Tested?	Commented?
DB-01	PostgreSQL schema design with Los Alamos Chess support	Yes	Yes	Yes
DB-02	Users table with authentication fields and constraints	Yes	Yes	Yes
DB-03	Game table with FEN string storage and game state tracking	Yes	Yes	Yes
DB-04	Game moves table (append-only) with immutability triggers	Yes	Yes	Yes
DB-05	Audit logging table for security and change tracking	Yes	Yes	Yes
Database Management & Operations				
DB-06	Connection pooling with environment-based configuration	Yes	Yes	Yes
DB-07	Schema deployment script (setup.js) with validation	Yes	Yes	Yes
DB-08	Database connectivity diagnostics (test-connection.js)	Yes	Yes	Yes
DB-09	Test data seeding script for development environments	Yes	Yes	Yes
DB-10	Performance indexes for users, games, and moves	Yes	Yes	Yes
Advanced Features & Data Integrity				
DB-11	Optimistic locking for concurrent game state updates	Yes	Yes	Yes
DB-12	Foreign key constraints and referential integrity	Yes	Yes	Yes
DB-13	Tournament and friendship system tables	Yes	Yes	Yes
DB-14	Data validation triggers and check constraints	Yes	Yes	Yes
DB-15	Refresh token management and idempotency table	Yes	Yes	Yes

Testing Infrastructure & Quality Assurance

Unit Test Suite: run-tests.js (orchestration), test-framework.js (custom framework), db-core-tests.js (user/game management), db-advanced-tests.js (constraints/performance)

Test Coverage: DB-01 through DB-14 comprehensive validation including user creation, game state management, move validation, constraint testing, performance benchmarks, and security features

Performance Metrics: Connection Setup < 100ms, User Creation < 50ms, Game Creation < 30ms, Move Addition < 20ms, Index Lookup < 10ms, Test Coverage 95%

Database Schema & Architecture

Core Tables: users (authentication & profiles), game (Los Alamos variant state), game_move (immutable history), audit_log (security tracking), friend (social features), tournament (competition system), idempotency (API safety)

Security Features: bcrypt password hashing (12 salt rounds), UUID primary keys, constraint-based validation, audit trail logging, SQL injection prevention via parameterized queries

Performance Optimizations: Strategic indexes on email/username lookups, game player queries, move history retrieval, connection pooling with configurable limits, query optimization with EXPLAIN ANALYZE

Demo Readiness & Integration Points

Functional: Complete schema deployment, user management with authentication, game state persistence, move history tracking, constraint validation, audit logging, test data seeding, connection diagnostics

Integration Ready: Byron's Rules Engine (game state validation), Elizabeth's Security Layer (user authentication), Ethan's WebSocket Server (real-time updates), Nastasha's Frontend (data requirements)

Code Metrics: Schema Definition 300 lines, Connection Management 150 lines, Setup Scripts 200 lines, Test Suite 800 lines, **Total 1,450 lines**

Status: 100% of assigned database tasks completed, production-ready with comprehensive testing and documentation

EPE Progress Demo Status - Security Section

Date: 22 September 2025 **Team Member:** Elizabeth Vorster - u22639170 **Role:** Security & Authentication Developer

Completed Tasks (Done Status)

Core Authentication System Implementation				
ID	Task	Completed?	Tested?	Commented?
SEC-01	JWT token generation and validation system	Yes	Yes	Yes
SEC-02	User registration endpoint with validation	Yes	Yes	Yes
SEC-03	User login endpoint with credential verification	Yes	Yes	Yes
SEC-04	Password hashing with bcrypt (12 salt rounds)	Yes	Yes	Yes
SEC-05	Rate limiting for authentication endpoints	Yes	Yes	Yes
Security Middleware & Authorization				
SEC-06	JWT token authentication middleware	Yes	Yes	Yes
SEC-07	Role-based authorization middleware	Yes	Yes	Yes
SEC-08	Input sanitization and XSS prevention	Yes	Yes	Yes
SEC-09	CORS configuration for frontend integration	Yes	Yes	Yes
SEC-10	WebSocket authentication for real-time features	Yes	Yes	Yes
Advanced Security Features				
SEC-11	Refresh token mechanism for session management	Yes	Yes	Yes
SEC-12	Password reset functionality	Yes	Yes	Yes
SEC-13	Audit logging for security events	Yes	Yes	Yes
SEC-14	Move authorization for game integrity	Yes	Yes	Yes
SEC-15	Comprehensive error handling and status codes	Yes	Yes	Yes

Code Quality & Integration

Unit Tests: auth-middleware.test.js (JWT validation, rate limiting, XSS prevention), auth.test.js (user registration, login flows, token management), websocket-auth.test.js (real-time authentication, socket security), security integration tests (end-to-end auth flows), performance stress tests (concurrent authentication load)

Code Structure: security/ (auth-middleware.js, auth-routes.js, websocket-auth.js, move-authorization.js, index.js), src/models/ (AuthenticationService.js), comprehensive middleware pattern with separation of concerns, modular authentication system design

Performance: JWT Token Generation < 10ms, Token Validation < 5ms, Password Hashing ~ 100ms (bcrypt), Rate Limit Check < 1ms, WebSocket Auth < 15ms, Test Coverage 95%

Demo Readiness & Metrics

Functional: Complete user registration and login system, JWT access/refresh token management with 15min/1hr expiry, Rate limiting (10 auth attempts/5min, 10 moves/5sec), XSS prevention through input sanitization, WebSocket authentication for real-time features, Role-based authorization middleware, Password security (bcrypt 12 rounds, strength validation), Comprehensive error handling with proper HTTP status codes

Integration Status: Frontend login/register forms working end-to-end, Database integration with Arno's PostgreSQL schema, WebSocket authentication integrated with Ethan's real-time system, Rules engine authorization for move validation, Audit logging for security events, CORS configuration for cross-origin requests

Code Metrics: AuthenticationService ~ 400 lines, Auth Middleware ~ 300 lines, Auth Routes ~ 350 lines, WebSocket Auth ~ 200 lines, Unit Tests ~ 600 lines, **Total ~ 1,850 lines** achieved.

EPE Progress Demo Status - Server Section

Date: 22 September 2025 Team Member: Ethan (u21501212) Role: Server & Networking Developer

Completed Tasks (Done Status)

Core Server Implementation				
ID	Task	Completed?	Tested?	Commented?
SRV-01	Express.js server setup with REST API endpoints	Yes	Yes	Yes
SRV-02	WebSocket integration with Socket.IO for real-time	Yes	Yes	Yes
SRV-03	Health check endpoint with uptime monitoring	Yes	Yes	Yes
SRV-04	Environment configuration with .env file	Yes	Yes	Yes
SRV-05	Graceful shutdown handling for SIGTERM/SIGINT	Yes	Yes	Yes
Authentication & Security				
SRV-06	JWT access/refresh token system (15m/7d expiry)	Yes	Yes	Yes
SRV-07	Rate limiting for API and login endpoints	Yes	Yes	Yes
SRV-08	Helmet security headers and CORS configuration	Yes	Yes	Yes
SRV-09	WebSocket authentication middleware	Yes	Yes	Yes
SRV-10	Input validation with Joi for API endpoints	Yes	Yes	Yes
Real-Time Features				
SRV-11	Game room joining and move broadcasting	Yes	Yes	Yes
SRV-12	Chat message broadcasting with sanitization	Yes	Yes	Yes
SRV-13	User online status tracking with disconnect delay	Yes	Yes	Yes
SRV-14	Comprehensive error handling for API/WebSocket	Yes	Yes	Yes
SRV-15	Logging system for server events and errors	Yes	Yes	Yes

Code Quality & Integration

Unit Tests: server.test.js (health check, auth endpoints), auth.test.js (registration, login, refresh), websocket.test.js (real-time events), integration.test.js (end-to-end flows)

Code Structure: backend/ (server.js, config/database.js, middleware/auth.js, utils/logger.js, utils/validation.js), modular middleware design, separation of concerns

Performance: Health Check < 5ms, Token Validation < 5ms, API Response < 20ms, WebSocket Auth < 15ms, Test Coverage 85%

Demo Readiness & Metrics

Functional: REST API (health, auth, profile), WebSocket events (join-game, make-move, chat-message), JWT auth with rate limiting (10 logins/15min, 100 API/15min), CORS, Helmet, graceful shutdown, logging

Integration Ready: Arno's DB (PostgreSQL pool), Elizabeth's security (JWT, bcrypt), Byron's rules engine (move validation), Nastasha's frontend (CORS, WebSocket)

Code Metrics: Server Logic ~ 500 lines, Middleware ~ 250 lines, Utilities ~ 300 lines, Tests ~ 350 lines, **Total** ~ 1,400 lines

Status: **All of assigned server tasks completed for the progress demo**, not entirely production-ready yet with alot of comprehensive testing and documentation done so far but some port conflicts need to be handled.

EPE Progress Demo Status - Frontend Section

Date: 22 September 2025 **Team Member:** Nastasha van der Sandt u20422327 **Role:** Frontend Developer

Completed Tasks (Done Status)

Core Frontend Implementation				
ID	Task	Completed?	Tested?	Commented?
FE-01	Dashboard page with navigation (Play, Tournament, Learn, Logout)	Yes	Yes	Yes
FE-02	Display of last 3 games (dummy values) on dashboard	Yes	Yes	Yes
FE-03	Game view with 6×6 board (Los Alamos variant)	Yes	Yes	Yes
FE-04	Pawn movement implementation with move history (white only)	Yes	Yes	Yes
FE-05	Login page (form input, submission, basic validation)	Yes	Yes	Yes
FE-06	Registration page (character limit validation enforced)	Yes	Yes	Yes
Secondary Features & Setup Pages				
FE-07	Tournament setup page (basic structure, no full logic)	Yes	Yes	Yes
FE-08	Tutorial page with setup instructions (static content)	Yes	Yes	Yes
FE-09	Logout functionality (navigation back to login)	Yes	Yes	Yes
Testing & Deployment				
FE-10	Yes testing of navigation across pages	Yes	Yes	Yes
FE-11	Yes validation of pawn movement and history display	Yes	Yes	Yes
FE-12	Cross-browser smoke test (Chrome, Edge)	Yes	Yes	Yes

Code Quality & Integration

Unit Tests: None written yet. Testing was conducted manually on the web frontend (navigation, forms, movement).

Code Structure: Organized by feature pages: `dashboard.html`, `game.html`, `login.html`, `register.html`, `tournament_setup.html`, `tutorial.html`. Shared JS utilities in `/common/` for UI and WebSocket placeholders.

Performance: Pages load instantly with lightweight assets (no external heavy libraries). Navigation between static pages is seamless.

Demo Readiness & Metrics

Functional: - Dashboard navigation to play, tournament, learn, and logout - Game board with basic pawn movements and white move history - Display of last 3 dummy game results - Login and registration system (basic validation with character limit) - Tournament setup and tutorial page (static view only)

Integration Ready: - Hooks in place for backend authentication and game state persistence - Dashboard prepared to pull real values once backend integration completes - Game board ready to extend to full piece movement rules

Code Metrics: HTML ~ 1500 lines, CSS ~ 1000 lines, JavaScript ~ 500 lines, **Total ~ 3000 lines**

Status: **Core frontend demo tasks completed and manually tested**, backend integration and automated test coverage still pending.